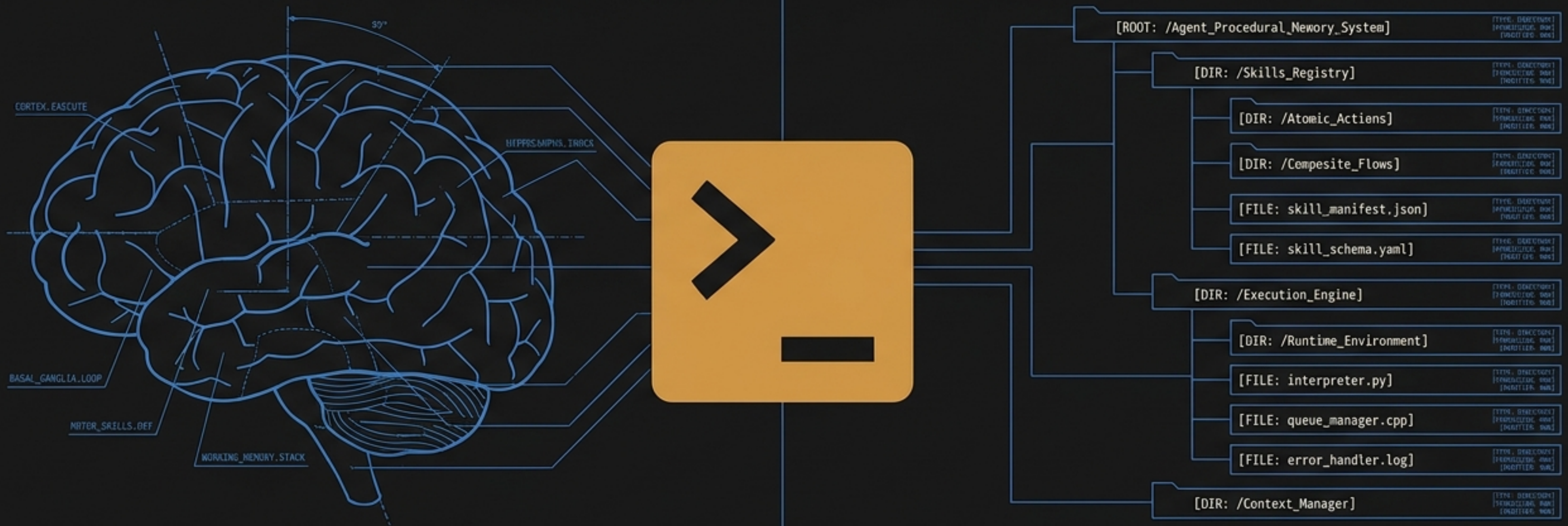


Engineering Procedural Memory

The anatomy, architecture, and execution of AI Agent Skills.



[STANDARD: AgentSkills Apache 2.0]

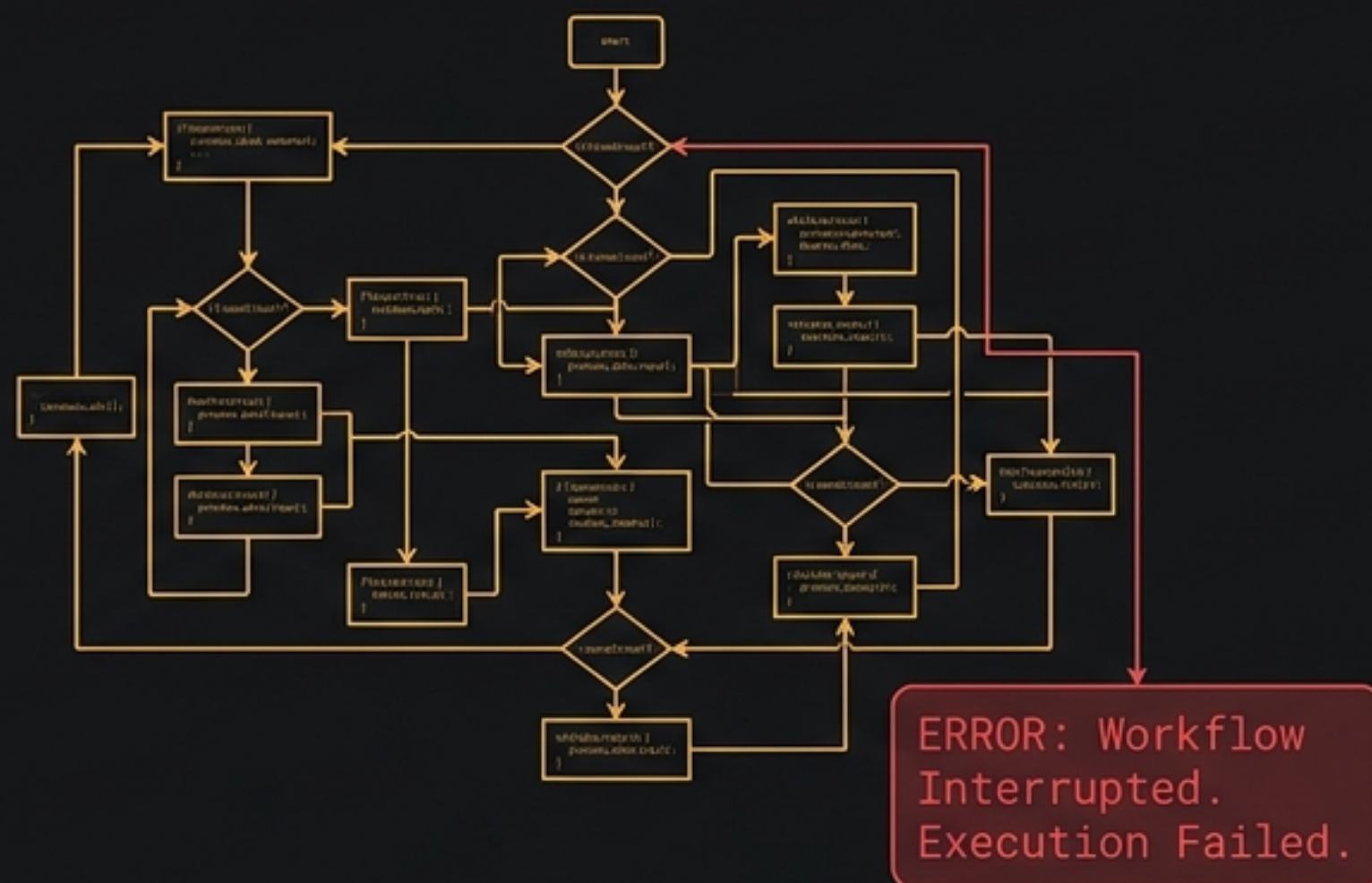
The LLM Baseline



- Kubernetes architecture
- The history of SQL
- The airspeed velocity of an unladen swallow

Agents have two options for workflows: be prompted with all 47 steps every single time, or guess. We need a third option.

The Missing Piece



- A 47-step workflow for generating a compliant financial report

Mirroring Human Cognitive Architecture

Semantic Memory

Fact-Based



Rome is the
capital of Italy.

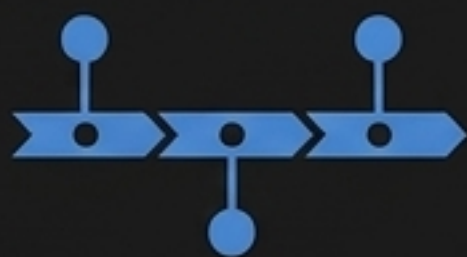
(Human)

(AI Mapping)

Retrieval Augmented
Generation (RAG)

Episodic Memory

Experience-Based



I went to Rome
last summer.

(Human)

(AI Mapping)

Conversational
Logs & History

Procedural Memory

Action-Based



How to ride a
scooter in Rome
and survive.

(Human)

(AI Mapping)

Agent Skills
Files

The Knowledge Paradigm Matrix

	Function	Medium	Permanence	Limitation / Advantage
RAG	Factual	Reference Material	Dynamic	Does not teach how to execute tasks.
Fine-Tuning	Domain Baseline	Model Weights	Permanent	Expensive, breaks on model updates.
MCP	Tool Access	External APIs	Runtime	Lacks judgment on when to use tools.
Skills	Procedural Knowledge	Markdown Files	Version Controlled	Provides the missing judgment and workflow.

The Elegance of the Open Standard

A comically simple format. A single file. Infinite workflows.



Apache 2.0 Licensed. Build once, deploy across any spec-compliant platform.

The Anatomy of a Procedural Memory File

YAML Frontmatter

```
---  
name: PDF Builder  
description: Use this when the user asks to extract a PDF.  
---
```

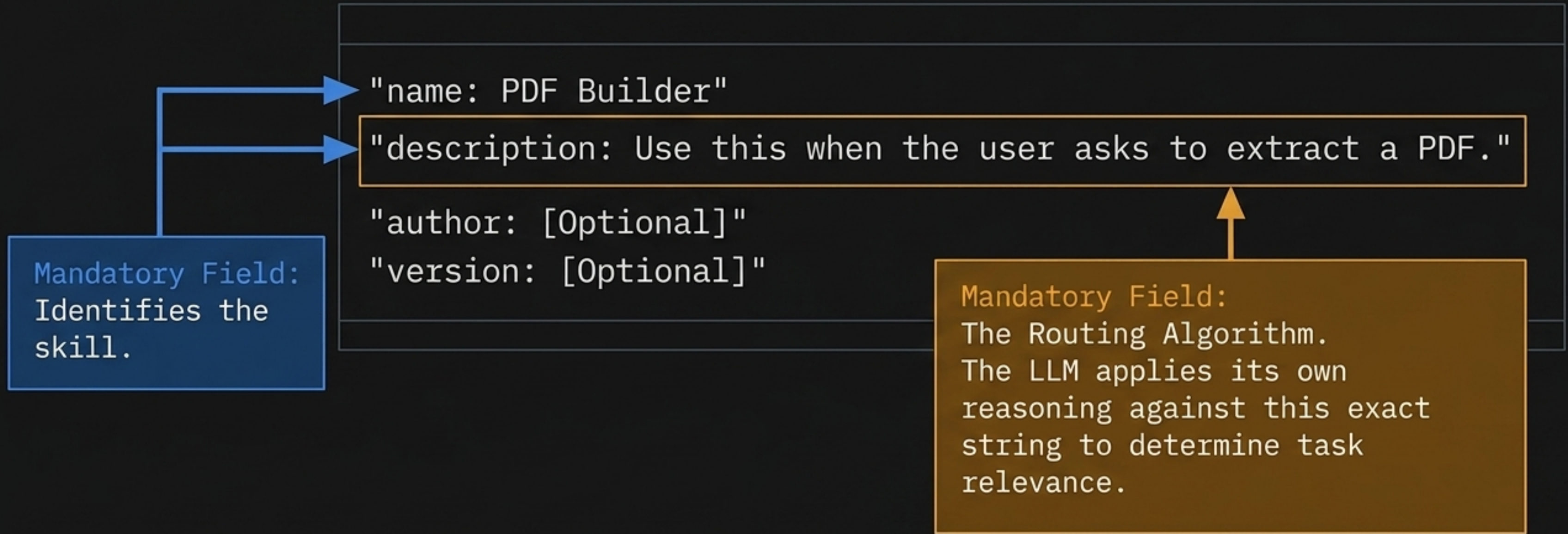
The Conditional Trigger:
Tells the agent exactly
WHEN to activate.

Plain Markdown

```
### Workflow  
  
1. Receive user document.  
2. Validate PDF format.  
3. Extract raw text structure.
```

The Execution Body:
Tells the agent exactly
WHAT to do.

Designing the Trigger Condition



⚠ A skill is only as powerful as the precision of its description.

Architecting the Execution Body

skill.md

Step-by-Step Workflow

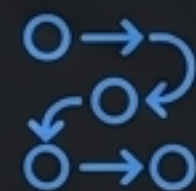
... (numbered steps would go here)

Strict Rules

... (rules would go here)

Examples (Input/Output)

... (input/output examples would go here)



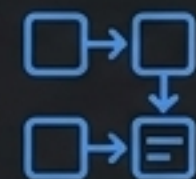
Workflows

Break procedures into deterministic, numbered steps to guide the LLM linearly.



Boundaries

Define strict operational rules and negative constraints the agent cannot violate.

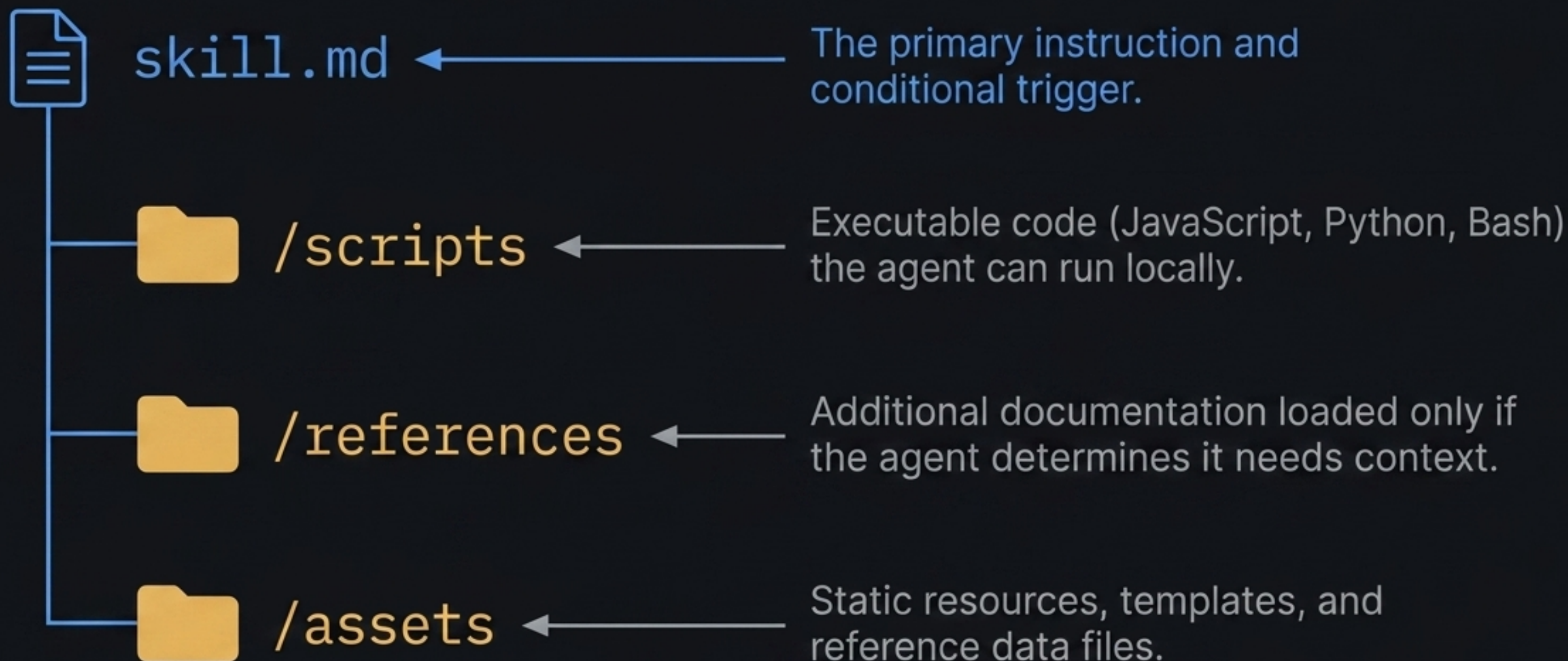


Few-Shot Mapping

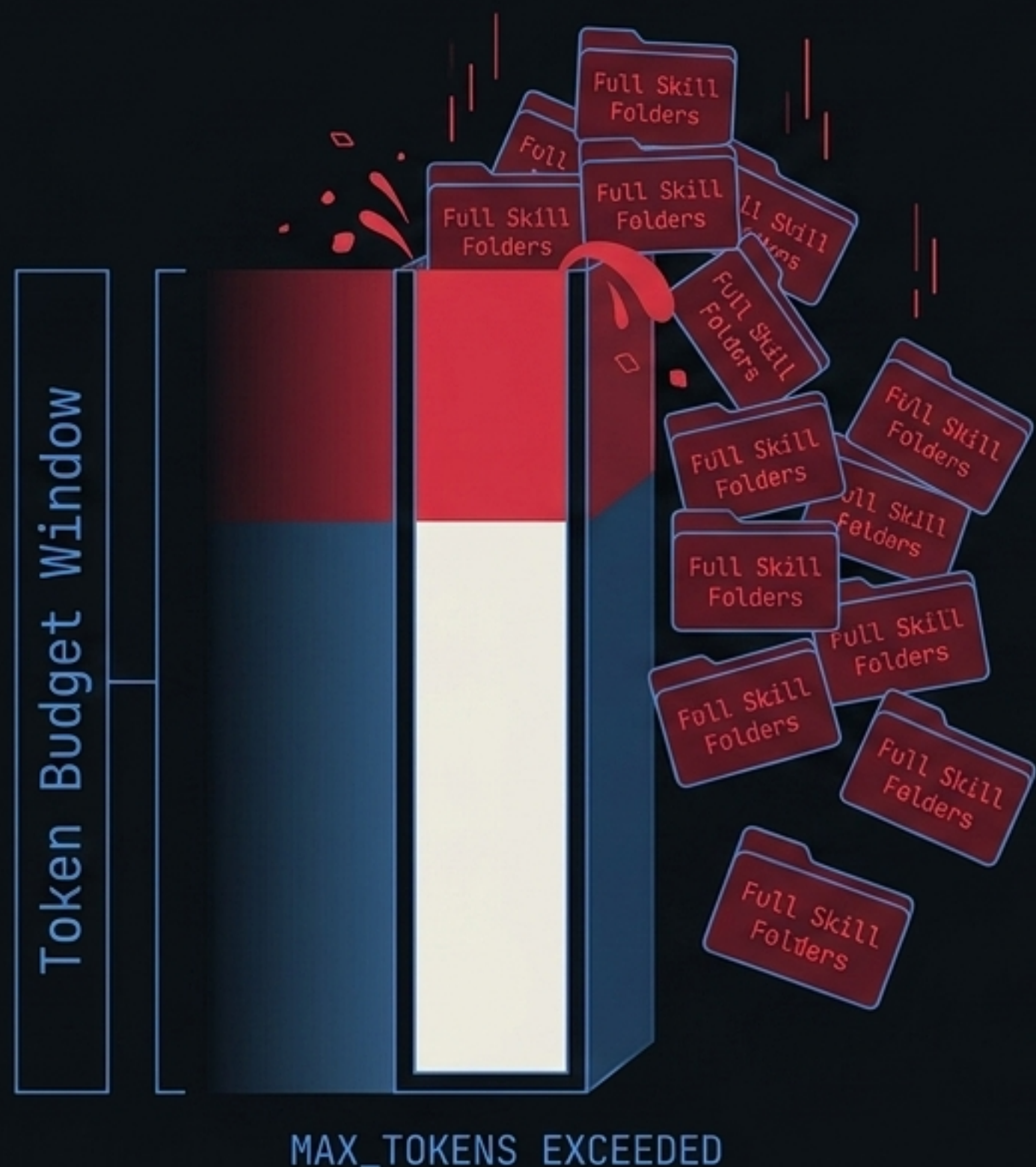
Provide clear input-to-output examples so the agent grasps the required formatting instantly.

The Master Folder Architecture

How to build a well-structured skills directory.



The Context Window Collision



The Scale Problem

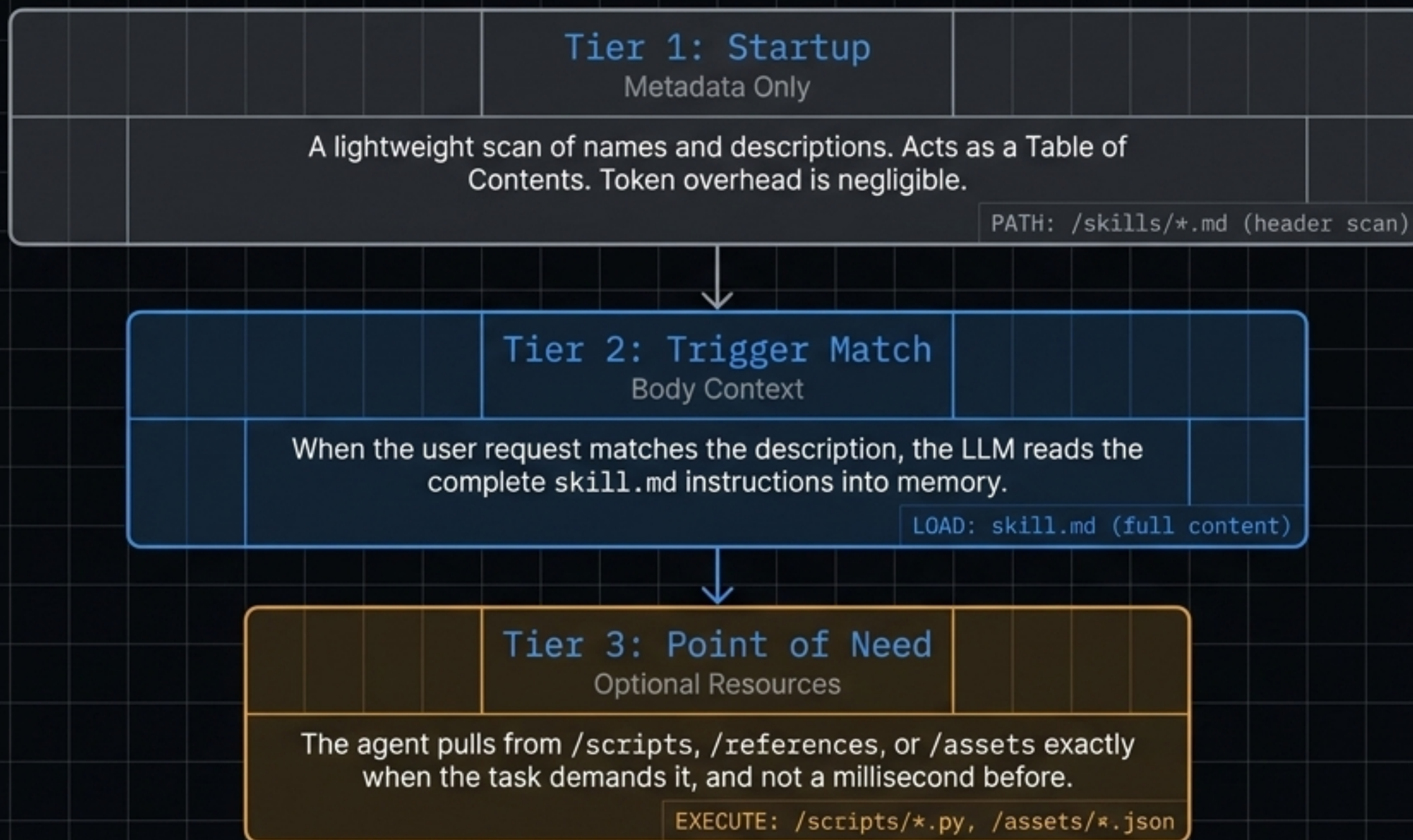
What happens when an agent has hundreds of defined skills?

The Outcome

Loading full instruction sets into the LLM context window at startup blows through the token budget before a single user question is asked.

The Engine: Progressive Disclosure

How the system intelligently loads resources exactly when needed.



The Execution Peril

```
> ./scripts/deploy.sh --override-security --key=$API_KEY
```

The Power: Skills in the /scripts directory have access to local file systems, environment variables, and API keys.

The Vulnerability Matrix

Prompt Injection

Manipulated
instructions.

Tool Poisoning

Hijacked API
pathways.

Hidden Malware

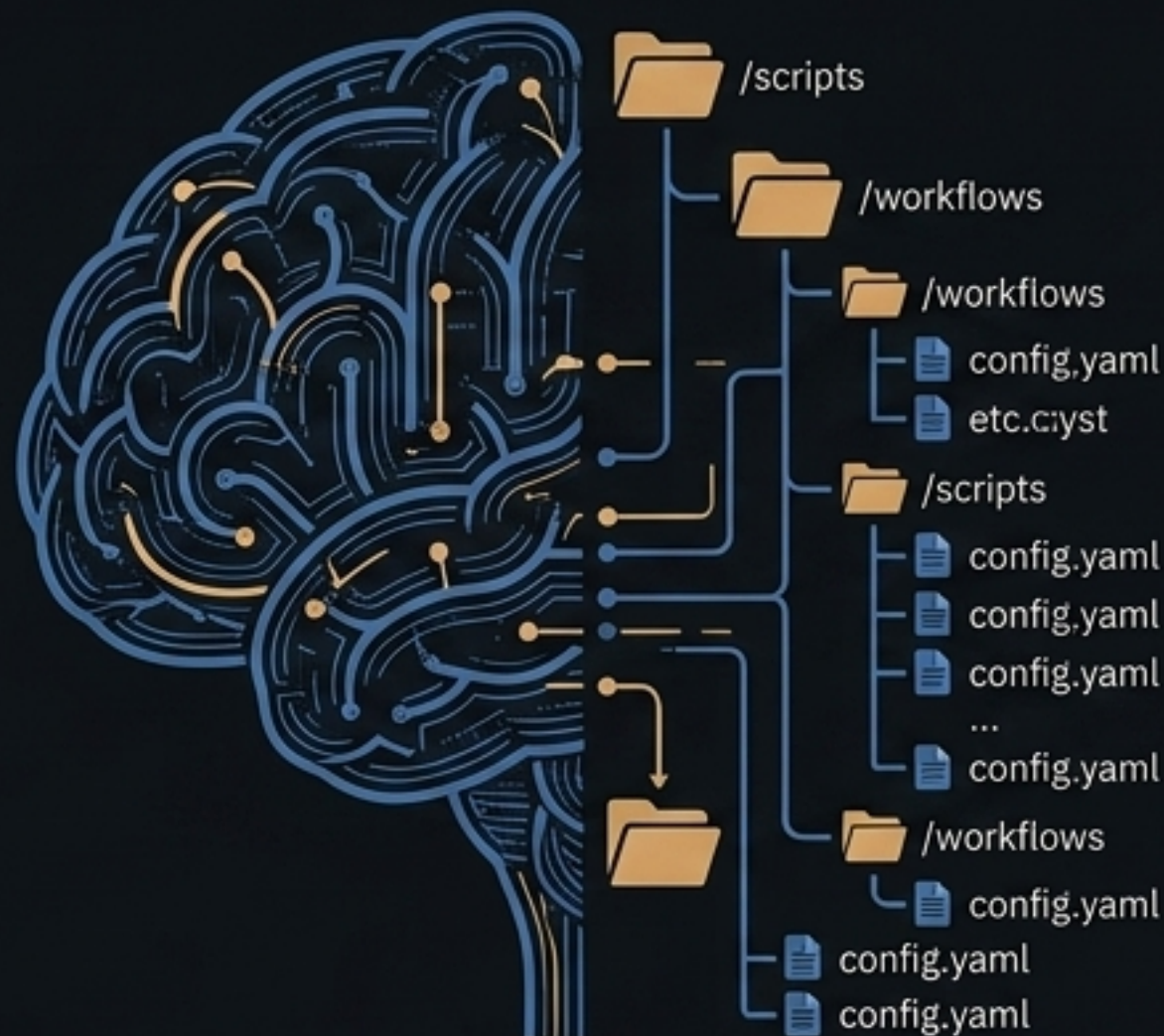
Malicious local
execution.

Core Philosophy: Treat skill installation the way any responsible team treats software dependencies.

- [x] **Audit** the `/scripts` directory before execution.
- [x] **Verify** the `routing` logic in the `YAML` description.
- [x] **Sandbox** local machine environments where possible.
- [x] **Review** `publicly` available `skills` for hidden payloads.

Procedural Memory, Mastered

- Defined in a simple markdown folder.
- Conditionally triggered via LLM reasoning.



- Loaded efficiently via progressive disclosure.
- Built on a universal, Apache 2.0 open standard.

Agents that know the facts can now learn the workflows. Teach them how.